

Bioinformatics

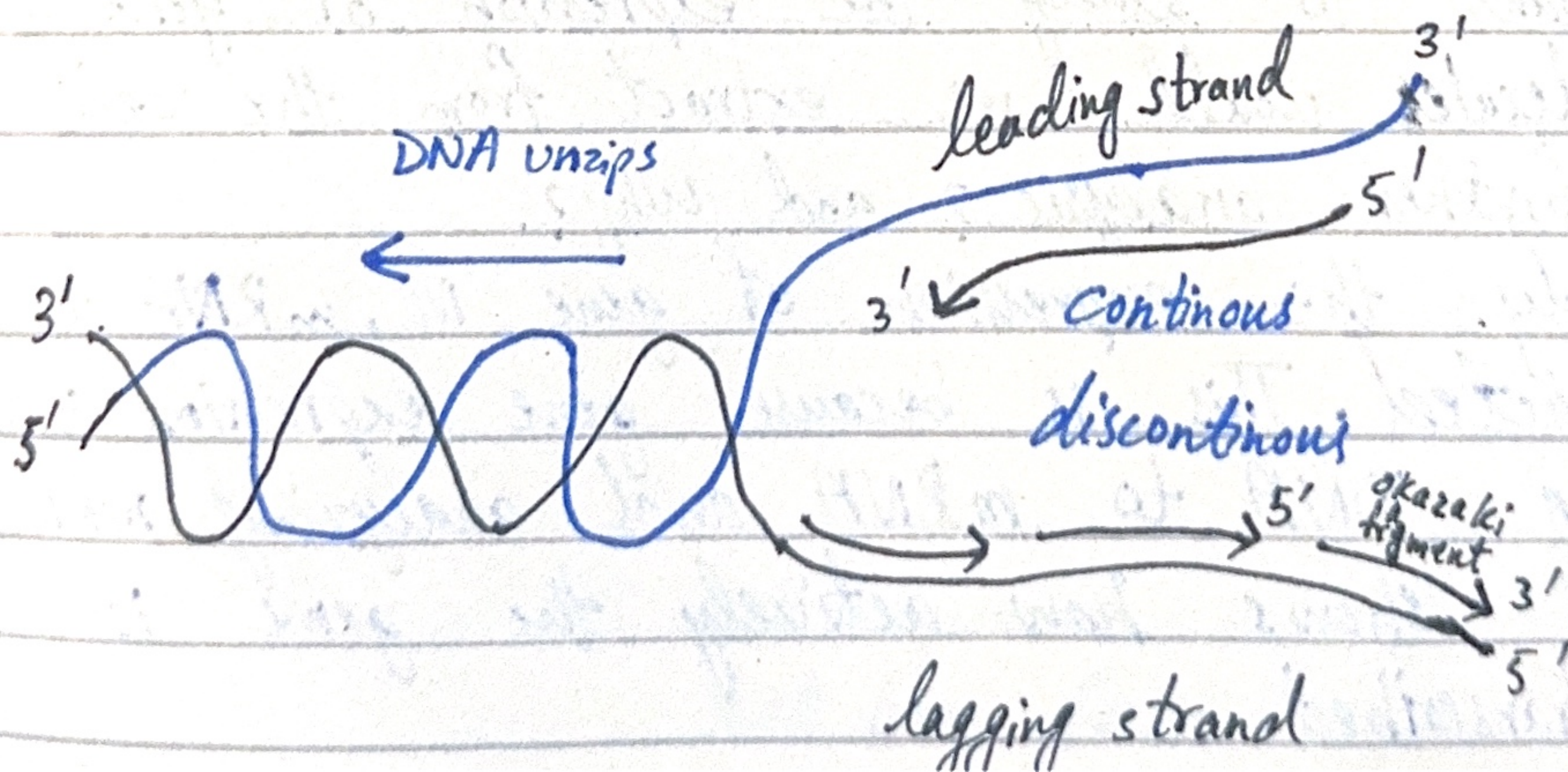
(Open Book Quiz - 1)

Q1) What is difference of open reading frame (ORF) and a Reading frame?

Ans) Open-Reading Frame : It is a continuous stretch of codons that start codon (AUG) and ends with a stop codon (UAA, UAG, or UGA). It has the potential to code for a protein.

1. Reading Frame : It is the way nucleotides in mRNA are grouped into codon (sets of three bases) during translation. There are three possible reading frames for a single ~~reversed~~ strand, depending on where translation starts.

Q2) Illustrate the complete process of replication?



Q-3) What is the third base wobble?

Ans) The third base wobble refers to the flexibility in pairing b/w the third base of a codon and the first base of the anti-codon in tRNA. Due to this wobble, some tRNAs can recognize multiple codons, reducing the number of tRNAs needed to decode all 61 codons.

Q-4) "Gene Expression" means?

Ans) Gene expression is the process by which information in a gene is used to produce a functional product, usually a protein. It involves two main steps:-

Transcription: (DNA $\rightarrow$ mRNA)

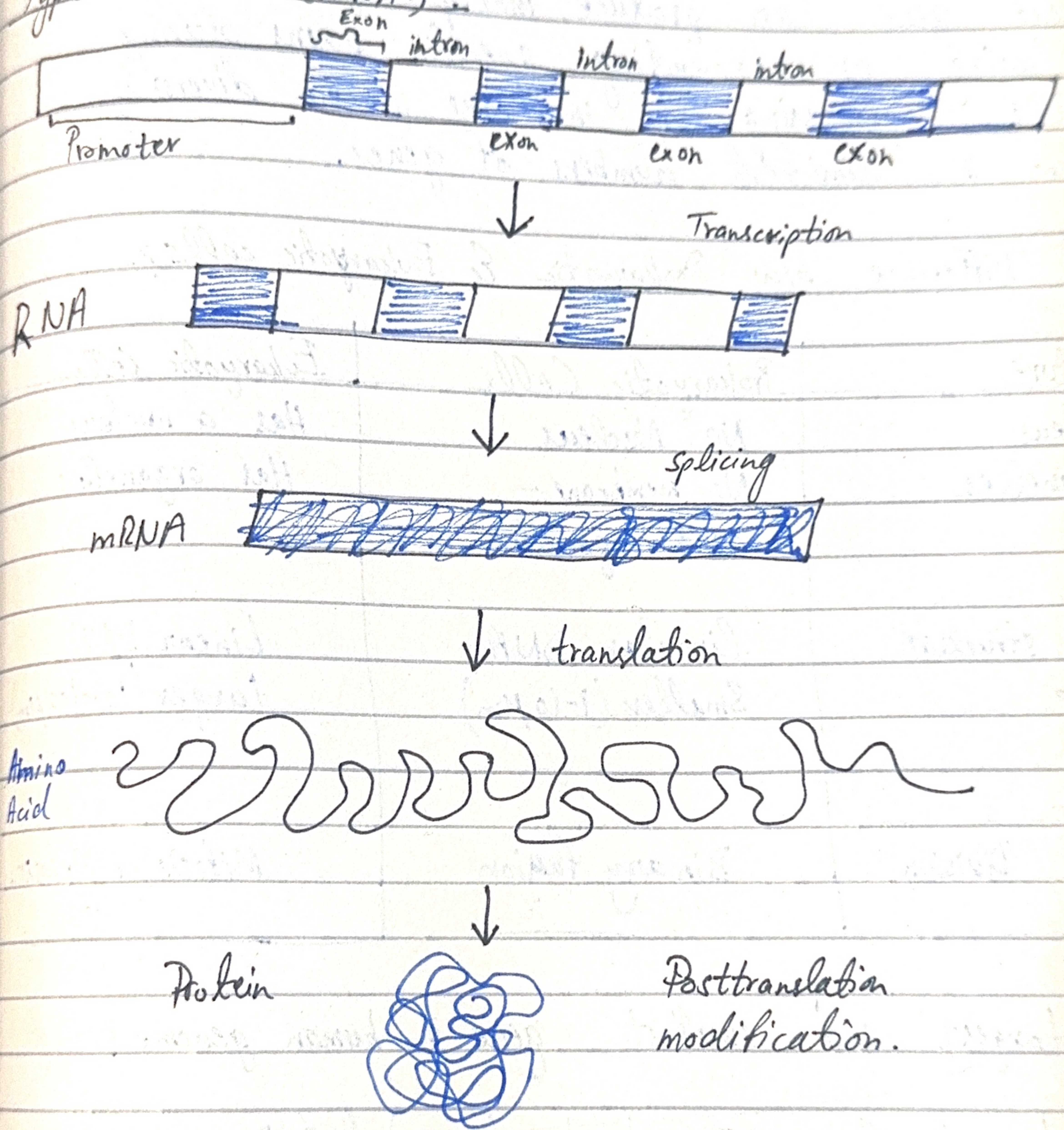
Translation: (mRNA $\rightarrow$ Protein)

Regulation of gene expression ensures that proteins are made only when needed.

Q-5) If we want to study the expression of gene M, what molecules would we extract from the cell for further analysis? and why?

Ans) To study the expression of gene M, mRNA would be extracted. This is because gene expression involves converting DNA to mRNA and analyzing mRNA levels shows how actively the gene is being transcribed.

Q-6) Structure of Typical gene and explain each feature?



- Promoter: Binding site for RNA polymerase, controls transcription starts.
- Exons: Coding regions that form final protein.
- Introns: Non-coding regions removed during splicing.
- Start Codon (AUG): Signals the beginning of translation.
- Stop Codon: Signals translation to stop.
- Poly-A-tail: Protects mRNA from degradation.

Q=7) ~~Alt~~ Alternative Splicing:-

It is a process where a single gene can produce multiple proteins by including or excluding specific exons during mRNA processing. This increases protein diversity from a limited numbers of genes.

Q=8) Difference b/w Prokaryotic & Eukaryotic cells?

Feature	Prokaryotic Cells	Eukaryotic Cells
Nucleus	No Nucleus	Has a nucleus
Organelles	No membrane-bound organelles	Has organelles
DNA structure	Circular DNA	Linear DNA
Size	Smaller (1-10 μm)	Larger (10-100 μm)
Size Cell Division	Binary fission	Mitosis or Meiosis

Q=9) Largest and smallest gene in human genome?

Largest Gene: Dystrophin (DMD), (~2.3 million base

Smallest Gene: Testis Determ pairs long)

Q=9) Largest and smallest gene in human genome?

Largest Gene: Dystrophin (DMD), (~2.3 million base pair long)

Smallest Gene: Testis determining factor (TDF), (around 14,000 base pairs)